

Alexander Belsten

Redwood Center for Theoretical Neuroscience
University of California, Berkeley
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EDUCATION

- Ph.D. **University of California, Berkeley**, Berkeley, California
Vision Science (2021-25)
Advisor: Bruno A. Olshausen
Dissertation: *Efficient coding of color in natural images*
- B.S. **Rensselaer Polytechnic Institute**, Troy, New York
Computer and Systems Engineering (2016-20)
GPA: 3.78

RESEARCH INTERESTS

Vision
Theoretical neuroscience

APPOINTMENTS

- 2026– **University of California, Berkeley**, Berkeley, California
Postdoctoral researcher, Redwood Center for Theoretical Neuroscience
Advisor: Bruno A. Olshausen
- 2021–25 **University of California, Berkeley**, Berkeley, California
Graduate student researcher, Redwood Center for Theoretical Neuroscience
Advisor: Bruno A. Olshausen
- 2021 **Washington University in St. Louis**, Saint Louis, Missouri
Research assistant & systems engineer, Department of Neurosurgery
Advisor: Peter Brunner
Improved and maintained BCI2000. List of contributions:
(belsten.github.io/research/#contributions-to-bci2000)
- 2020–21 **Rensselaer Polytechnic Institute**, Troy, New York
Research assistant, Intelligent Structural Systems Laboratory
Advisor: Fotis Kopsaftopoulos
- 2018–21 **National Center for Adaptive Neurotechnologies**, Albany, New York
Research assistant
Advisors: Peter Brunner, Gerwin Schalk

PUBLICATIONS

Citation data available at **Google Scholar**

- 2026 Z. Yun, A. Belsten, D. Bi, Z. Kadhodaie, Y. Chen, and B. A. Olshausen. "Toward a mechanistic understanding of inference in visual cortex and diffusion models." arXiv:2607.15693 [q-bio.NC].
- 2026 A. Belsten, E. P. Frady, and B. A. Olshausen. "Emergence of unique hues from sparse coding of color in natural scenes." *Journal of the Optical Society of America A* 43 (8), 1210-1223.
- 2025 A. Belsten and B. A. Olshausen. "Emergence of strategic cone weighting from efficient coding of spatiochromatic natural images." *Journal of the Optical Society of America A* 42 (5), B495-B502.
- 2024 P. Demarest, N. Rustamov, J. Swift, T. Xie, M. Adamek, H. Cho, E. Wilson, Z. Han, A. Belsten, N. Luczak, P. Brunner, S. Haroutounian, E. C. Leuthardt. "A novel theta-controlled vibrotactile brain-computer interface to treat chronic pain: A pilot study." *Scientific Reports* 14 (1), 3433.
- 2022 C. Bybee, A. Belsten, and F. T. Sommer. "Cross-frequency coupling increases memory capacity in oscillatory neural networks." arXiv:2204.07163 [q-bio.NC].
- 2022 G. Schalk, S. Worrell, F. Mivalt, A. Belsten, I. Kim, J. M. Morris, D. Hermes, B. T. Klassen, N. Staff, S. Messina, T. Kaufmann, J. Rickert, P. Brunner, G. Worrell and K. J. Miller. "Towards a fully implantable ecosystem for adaptive neuromodulation in humans." *Frontiers in Neuroscience* 16, 932782.
- 2021 A. Belsten and F. Kopsaftopoulos. "Data-driven flight state identification via time-series-informed features and convolutional neural network." *AIAA Aviation Forum*, 3182.
- 2020 A. Belsten, M. Adamek, and P. Brunner. "Hardware abstraction to facilitate the dissemination and validation of electrophysiological experiments." *IEEE EMBS*.

INVITED TALKS

- 2026 **Energy Consequences of Information Workshop**, Santa Fe, New Mexico
- 2025 **Vision Brunch**, Stanford University, Stanford, California
- 2025 **Optica Webinar: Color Data Blast for Early-Career Researchers**, Virtual
- 2025 **Vision Sciences Society Meeting (VSS)**, St. Pete Beach, Florida

CONFERENCE POSTERS

- 2026 **Computational and Systems Neuroscience (COSYNE)**, Lisbon, Portugal
- 2025 **Society for Neuroscience (SfN)**, San Diego, California
Presenting author on one poster; co-author on two others
- 2025 **Computational and Systems Neuroscience (COSYNE)**, Montreal, Canada
- 2024 **International Color Vision Society Meeting (ICVS)**, Ljubljana, Slovenia
- 2023 **Society for Neuroscience (SfN)**, Washington D.C
Presenting author on one poster; co-author on one other
- 2022 **Bay Area Vision Research Day (BAVRD)**, University of California, Berkeley, Berkeley, California
- 2022 **Computational and Systems Neuroscience (COSYNE)**, Lisbon, Portugal
- 2021 **Society for Neuroscience (SfN)**, Virtual
Presenting author on one poster; co-author on two others
- 2021 **NIH BRAIN Initiative**, Virtual

2020 **Society for Neuroscience (SfN)**, Virtual
Presenting author on one poster; co-author on one other

GRANTS AND AWARDS

2025 **National Eye Institute Early Career Scientist Travel Grant**

2016 **Rensselaer Leadership Award**

TEACHING EXPERIENCE

2022 **Neural Computation** (VS265), University of California, Berkeley
Graduate student instructor

2020 **Digital Signal Processing** (ECSE4530), Rensselaer Polytechnic Institute
Undergraduate teaching assistant

2018 **Data Structures** (CSCI200), Rensselaer Polytechnic Institute
Undergraduate mentor

2018 **Foundations of Computer Science** (CSCI2200), Rensselaer Polytechnic Institute
Undergraduate mentor

SERVICE AND LEADERSHIP

2022– **Active contributor**, Sparse Coding Repository
(github.com/rctn/sparsecoding)

2022–23 **Social chair**, Berkeley Vision Science Student Government

2022 **Speaker committee**, Bay Area Vision Research Day

2019–20 **President (2019); webmaster (2020)**, IEEE-HKN honor society, Beta Nu Chapter

ACADEMIC HONORS

2016–20 **Rensselaer Dean's Honor List** (8 semesters)